

Chapter 156 — Pathology of Prostatic Neoplasia

PROSTATIC INTRAEPITHELIAL NEOPLASIA

- **Prostatic intraepithelial neoplasia (PIN)** = architecturally benign acini or ducts lined by **cytologically atypical cells**; subclassified into low-grade PIN (LGPIN) and high-grade PIN (HGPIN).
- **LGPIN is not reported**: pathologists cannot reproducibly separate it from benign tissue, and it carries **no excess cancer risk** on repeat biopsy compared with a benign biopsy.
- **HGPIN as precursor** — evidence:
 - **More and larger HGPIN foci** in prostates with cancer; more HGPIN → more multifocal carcinoma.
 - **Shared biomarkers and molecular changes**: about 20% of HGPIN carries the **TMPRSS2:ERG fusion** (transmembrane protease serine 2 : ETS-related gene), present in about 50% of prostate cancers.
 - **Direct antecedent** of adenocarcinoma, but **additional genomic alterations** are needed for invasion; carcinoma glands may be seen **budding off** HGPIN.
- **Incidence on biopsy about 4–5%**; reported range 0–25%, the spread reflecting **interobserver threshold** between LGPIN and HGPIN.
- **Unifocal HGPIN**: repeat-biopsy cancer risk **not significantly higher** than after a benign biopsy.
- **Multifocal HGPIN**: older series advised repeat biopsy; modern multiparametric magnetic resonance imaging (mpMRI)-guided re-sampling shows **very low risk of unsampled significant cancer** once digital rectal examination (DRE) and prostate-specific antigen (PSA) are considered.
- **Repeat biopsy is not indicated solely for isolated HGPIN**, unifocal or multifocal.
- HGPIN is less worrisome now because **cribriform HGPIN is no longer diagnosed** — it is reclassified as intraductal carcinoma or atypical intraductal proliferation (AIP).
- **HGPIN on transurethral resection (TUR)**: significance unclear (few, conflicting studies); on current data it **should not prompt further sampling** either.
- **PINATYP**: HGPIN with adjacent small atypical glands — outpouchings or tangential sections cannot be separated from infiltrating carcinoma (management: see table).

Intraductal Carcinoma and Atypical Intraductal Proliferation

- **Intraductal carcinoma of the prostate (IDC-P)**: architectural or cytological atypia **clearly exceeding HGPIN**; a more important lesion than HGPIN.
- **Prevalence rises with risk**: from about 2% in low-risk to over half in metastatic or recurrent cohorts; about 60% after androgen deprivation therapy (ADT) or chemotherapy.
- **IDC-P without invasive carcinoma on biopsy is very rare** (<0.06% of specimens).
- **Nature**:
 - **Usually** = **retrograde extension of high-grade invasive carcinoma** into ducts and acini, clonally related to adjacent invasive tumour; almost always accompanied by invasive, typically high-grade, carcinoma.
 - **Rarely** (radical prostatectomy without adjacent high-grade carcinoma) = a **high-grade precursor lesion**.
- **Prognosis**: on biopsy, **independently predicts early biochemical recurrence and metastatic failure after radiotherapy** (intermediate/high risk); enriched in **radio-recurrent** cancer (with cribriform); **adverse even when metastatic**.
- **IDC-P alone on biopsy**: some clinicians re-biopsy first to document invasion. **With grade group 1**, radical prostatectomy upgrades in >80%.

- **Guo–Epstein criteria** for IDC-P on biopsy are **stringent and highly specific**, which is what licenses definitive therapy without invasive carcinoma.
- **AIP**: the significant proportion of intraductal lesions **short of these criteria**; with a benign or grade group 1 biopsy, re-biopsy seeks IDC-P or significant cancer.

PRECURSOR AND INTRADUCTAL LESIONS COMPARED

Lesion	Defining feature	Relationship to invasive cancer	Action on isolated biopsy finding
LGPIN	Mild cytological atypia; irreproducible	No excess risk	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	Precursor; needs further genomic hits	No repeat biopsy (unifocal or multifocal)
PINATYP	HGPIN plus adjacent small atypical glands	Risk as for "atypical"	Manage as atypical, suspicious for carcinoma
AIP	Atypia beyond HGPIN, below IDC-P criteria	May herald IDC-P or significant cancer	Consider repeat biopsy
IDC-P	Atypia clearly beyond HGPIN (e.g. dense cribriform), basal cells retained	Usually spread of high-grade invasive cancer; rarely precursor	Definitive therapy (or re-biopsy); excludes active surveillance

- **What the table tests**: a gradient of atypia maps onto a gradient of action, "do not report" to "treat". **Trap**: re-biopsy for HGPIN; surveillance for grade group 1 with IDC-P.

Figure 156.1

- **(A) IDC-P with dense cribriform formation**; **(B) the same glands surrounded by basal cells** on immunostaining — the retained basal layer marks the process as intraductal.

Concept — *The intraductal lesions are graded by what they signify, not by what they look like: HGPIN is a weak marker of risk, IDC-P is usually a marker of high-grade cancer already present, and management follows that asymmetry.*

ADENOCARCINOMA

Staging Classification

STAGE DEFINITIONS USED IN THE PATHOLOGY OF PROSTATIC ADENOCARCINOMA

Stage	Definition
T1a	Incidental on TUR of the prostate (TURP) or enucleation; $\leq 5\%$ of tissue involved
T1b	Incidental on TURP or enucleation; $> 5\%$ involved
T1c	Non-palpable, found on needle biopsy (usually for raised PSA)
pT2	Organ-confined at radical prostatectomy (T1a–c are restaged as pT2 or pT3)
pT2+ (pT2x)	Organ-confined elsewhere, but positive margin at an intraprostatic incision or where the boundary is ill-defined (e.g. apex) — stage at that site cannot be assessed
pT3a	Extraprostatic extension (EPE) without seminal vesicle invasion ; includes microscopic bladder neck invasion
pT3b	Seminal vesicle invasion

- **Trap**: microscopic bladder neck invasion is **pT3a, not pT4**; T1a/T1b hinge on the **5% cut-off**, not on grade.

Location

- **Peripheral zone, posterior**: the main tumour mass in clinical T2 and in 85% of T1c tumours.

- **Anterior tumours** (transition zone or anterior horn of peripheral zone) in about 15% of radical prostatectomies: **same biochemical recurrence risk** as posterior tumours, but with EPE **more likely to have positive margins**.
- **Multifocal in >85%**.
- **Bilateral cancer**: **contralateral tumour usually small**, yet about 20% show contralateral adverse size, EPE, grade or margins.

Spread of Tumour

- **No discrete histological capsule** → "extraprostatic extension", not "capsular penetration".
- **Peripheral tumours exit via perineural space invasion**; EPE is **posterior and posterolateral**, mirroring tumour location.
- **Seminal vesicle invasion** = tumour in the **muscular wall** of the seminal vesicle. Routes, commonest first:
 - **Commonest**: exit at the base → periseminal vesicle soft tissue → seminal vesicle.
 - **Less common**: along ejaculatory ducts; direct extension from the base into the vesicle wall.
 - **Almost never**: discontinuous metastasis to the seminal vesicle.
- **Rectal involvement** is rare and may mimic a **rectal primary**.
- **Metastases**: commonest sites **lymph nodes and bone**; may present in **left supraclavicular nodes**.
- **Lung**: very common at autopsy, **almost always with bone**, usually **asymptomatic**; small nodules or lymphatic spread, not large deposits. **Then liver and adrenal**.

Tumour Volume

- **Size correlates with stage**: EPE uncommon <0.5 cm³; nodes or seminal vesicle invasion uncommon <4 cm³. **Volume tracks grade**; location and grade modulate its effect.
- **Transition zone tumours exit at larger volumes** than peripheral zone tumours — lower grade and farther from the edge of the gland.

Grade

- **Gleason system** (late 1960s–early 1970s): **glandular architecture at low magnification**, patterns 1 (most differentiated) to 5 (undifferentiated); substantially modified since (2005 and later).
- **Scoring rule**: original system = commonest + second commonest; **2005 update, on biopsy = commonest + highest pattern in a core**. **Single pattern** → doubled (e.g. 3+3).
- **Lowest score in practice is 3+3=6**, not 2 — so patients wrongly read "6 of 10" as mid-range aggressiveness.
- **Expanded pattern 4** in the modified system reclassified many former score 6 tumours to 7, so **modern score 6 is more homogeneous and more favourable**:
 - Virtually **no progression** of organ-confined score 6 after radical prostatectomy.
 - **Pure score 6 has no capacity for pelvic nodal metastasis**.
 - **EPE uncommon** with 3+3=6 at radical prostatectomy.
- **Grade group system**: five tiers, proposed at Johns Hopkins, validated multi-institutionally; **reflects biology better than Gleason score**. Validated for biochemical recurrence after radiotherapy, distant metastasis and cancer-specific mortality; endorsed by **WHO 2016 (World Health Organization)**, **CAP (College of American Pathologists)** and **AJCC (American Joint Committee on Cancer)**.

BOX 156.1 GRADE GROUPS DEFINITIONS — WITH PROGNOSTIC CORRELATES FROM THE CHAPTER

Grade group	Gleason score	Point of the tier	Nodal metastasis at radical prostatectomy	5-year biochemical recurrence-free survival
1	3+3=6	Indolent; "1" signals low risk as in other grading systems; no subdivision of well-formed glands needed	0%	96%
2	3+4=7	Very good prognosis, rare metastasis	0.6%	88%
3	4+3=7	Forces separation from 3+4 (no lumping as "7")	4.3%	63%
4	4+4=8; 3+5=8; 5+3=8	Forces separation of 8 from 9–10	—	48%
5	4+5=9; 5+4=9; 5+5=10	Highest grade	—	26%

- **Pattern:** the steep drop is **between grade groups 2 and 3**, so 3+4 and 4+3 are never merged. **Trap:** 3+5 and 5+3 are **grade group 4**; group 5 starts at 9.
- **Controversy:** pure grade group 1 cannot metastasise, so whether to call it "cancer" is debated.

Figure 156.2 — Gleason patterns and grade-group morphology

- **Discrete well-formed glands**, whatever their size or packing → **pattern 3** (grade group 1).
- **Cribriform** (including glomeruloid), **poorly formed and fused glands** → **pattern 4**.
- **Sheets, cords, nests, single cells, solid nests, comedonecrosis** → **pattern 5**.

Upgrading and reclassification

BIOPSY-TO-PROSTATECTOMY UPGRADING AND RECLASSIFICATION ON SURVEILLANCE

Setting	Risk of higher grade
Biopsy grade group 1 → radical prostatectomy	About one-third upgraded
Biopsy grade group 2 → radical prostatectomy	About 20% become grade group 3
Very low-risk grade group 1 on surveillance (1–2 cores, no core >50%, PSA density <0.15)	Close to 15% higher grade on later biopsy
Grade group 1 with negative mpMRI (no lesion or PI-RADS 1–2; Prostate Imaging–Reporting and Data System)	Still about 20% actuarial upgrading at 6 years
Grade group 1, yearly surveillance biopsies, first 3 years	Relatively low (about 19%) reclassification

- **Predictors of upgrading:** greater cancer extent on biopsy, **higher PSA, smaller prostate, fewer cores**.
- **Mechanism:** **sampling error** is the unavoidable cause. Early reclassification (within 3 years) mostly = **missed higher-grade component**, not progression; later change may be **true dedifferentiation or a new high-grade focus**.
- **Trap:** a negative mpMRI does not remove the risk of upgrading in grade group 1.

Concept — *The modern Gleason system was re-drawn so that score 6 means biologically indolent; grade groups simply relabel that biology so the numbers say what they mean, and sampling error, not tumour change, explains most apparent grade drift.*

Assessment of Needle Biopsy Specimens

Processing

- **Separate or ink-designated cores → a Gleason score per core; undesignated pooled cores → one overall score**, as if a single long core.

BOX 156.2 REASONS TO SUBMIT NEEDLE CORES IN SEPARATE JARS FOR EACH SEXTANT SITE

Benefit	How
Targeted re-biopsy	Atypical site can be preferentially re-sampled, plus other sites
Targeted prostatectomy sampling	Known cancer site guides extra blocks when initial sections show no cancer
Avoiding diagnostic pitfalls	Base: seminal vesicle or central zone mimics HGPIN ; apex: Cowper gland mimics cancer
Brachytherapy planning	Guides placement of extra seeds
No buried foci	Maximum 2–3 cores per block , fully visualised
No fragmentation	Preserves count of positive cores, % involvement and highest grade

- **What the box is for:** site-specific submission turns the pathology report into a map. **Trap:** pooled cores lose per-core grade and extent.

Differential Diagnosis

- **Under-diagnosis of limited cancer** is among the commonest problems in prostate pathology; benign mimickers abound.
- **Basal cell markers** — high-molecular-weight cytokeratin (HMWCK) and **p63** — label benign glands; **absence supports cancer**. **Pitfall:** partial atrophy, adenosis and HGPIN may also lack basal cells.
- **α-methylacyl-CoA racemase (AMACR)** preferentially labels carcinoma and HGPIN — an **adjunct only**, with common false positives and negatives.
- **"Atypical, suspicious for carcinoma":** about 5% of biopsies.
 - **Cancer on follow-up about 40%; PSA does not stratify** the risk.
 - About 80% of subsequent cancers are grade group 1; about 5% grade groups 4–5.
 - **Follow-up warranted:** serum or urine tests, imaging, or re-biopsy with **extra sampling of the atypical site**; only about 40% are re-biopsied.
 - **Atypical is the diagnosis most often changed on expert review → seek expert consultation before re-biopsy.**

Prognosis

- **Adverse biopsy findings predict adverse prostatectomy findings; favourable ones do not** (sampling error).
- **Extent measures:** number of positive cores, total mm of cancer, % per core, total % of specimen — **none convincingly superior**. Report **number of positive cores plus one other measure**.
- **Biopsy grade + clinical stage + PSA** together predict extent of disease more accurately.
- **Perineural invasion on biopsy** (report it): higher EPE; adverse on active surveillance; prognostic with **external-beam radiotherapy, less so brachytherapy**; predicts bone metastasis after radiotherapy plus ADT in locally advanced disease.
- **Atrophy** (all types): linked to carcinogenesis with inflammation, but common on biopsy and **does not raise cancer risk on re-biopsy**.

- **Gleason score is probably the most powerful biopsy predictor:** pathological stage; side-specific EPE and EPE into the neurovascular bundle; progression after radical prostatectomy; brachytherapy candidacy; radiotherapy prognosis; active surveillance entry and intervention; outcome after cryotherapy or high-intensity focused ultrasound (HIFU); focal therapy candidacy.
- **Percent pattern 4 within score 7** correlates with adverse pathology and biochemical recurrence → **report % pattern 4 for grade groups 2–3:**
 - **Grade group 2 with limited pattern 4** → may suit active surveillance (with age, comorbidity, imaging, extent, preference).
 - **Grade group 3 at 60% pattern 4** (near grade group 2) → neoadjuvant hormone therapy with radiotherapy **possibly avoidable**; **at 90%** (near grade group 4) → **recommended**.

Concept — *Every biopsy rule follows from sampling error being one-directional: a bad biopsy is believed, a good biopsy is provisional, and an atypical biopsy is re-examined before it is repeated.*

Assessment of Transurethral Resection Specimens

Processing

- **Staged by percentage of tissue involved** (5% cut-off; see staging table).
- **6–8 cassettes detect all T1b; 8–10 cassettes detect >90% of T1a.**
- Some institutions **embed all TUR tissue in men <65 years**, in whom aggressive therapy for T1a might be pursued.

Differential Diagnosis

- **Adenosis** — the chief mimic of **low-grade** adenocarcinoma: about 1.6% of benign TURs, 0.8% of needle biopsies; **transition zone**, often **multifocal**, incidental in TURs for obstruction.
- **Adenosis confers no increased risk** of harbouring or developing cancer.

Assessment of Radical Prostatectomy Specimens

Processing

- **Partial sampling protocols give accurate staging** where specimens are not totally embedded.
- **Whole-mount sections** look better for teaching and publication, but give **identical information** to routine sections.

Prognosis

Gleason score.

- **Grade each major nodule separately;** the largest (**dominant nodule**) usually has the highest stage and grade. If a smaller nodule is higher grade or stage, label nodules #1, #2 and grade each.
- **Radical prostatectomy grade correlates well with prognosis** (5-year biochemical recurrence-free survival by grade group: see Box 156.1 table).
- **Minor tertiary pattern 5** (<5% of a 3+4 or 4+3 nodule): **raises biochemical recurrence risk** (magnitude disputed — intermediate between grade groups 2 and 3, or equal to 3), but stays in its grade group because it **does not raise nodal risk**.
- **Percent pattern 4 within score 7** also adds prognostic information at prostatectomy.

Lymph node metastases.

- **Incidence has fallen** with screening; predicted by **clinical stage, PSA and biopsy grade**, so some surgeons omit lymphadenectomy at low risk.
- **Intraductal or cribriform carcinoma** at prostatectomy → higher risk of nodal and distant metastasis.

- **Uncommon with Gleason score <8** (rates in Box 156.1 table).
- **Node-positive disease almost always progresses** (occult distant metastasis). At 15 years: biochemical recurrence-free about 7%, metastasis-free about 42%, cancer-specific survival about 58%.
- **Predictors** after node-positive prostatectomy: **prostatectomy Gleason score and number and percentage of positive nodes**.

Extraprostatic extension and seminal vesicle invasion.

- **Capsule poorly defined, especially apex, base and anteriorly** → "EPE", not capsular invasion or penetration.
- Hard to recognise: vague boundaries, **desmoplasia** in periprostatic fat.
- **Focal EPE** (a few glands) **vs nonfocal EPE** (more extensive): degree **correlates with progression**; objective measures agree with it, none preferred.
- **Seminal vesicle invasion is far more ominous: 65% 5-year progression** after surgery.

Margins.

- **Positive margins** average about 15% in robotic series; **only about 50% progress** — re-excised tissue often shows no tumour.
- **Neurovascular-bundle frozen section** (resect the bundle if positive): described, **not routine** — time, staffing, false results, and other parameters suffice.
- **Distance from tumour to ink does not affect recurrence risk.**
- **Report margin extent and grade at the margin** — these guide **adjuvant radiotherapy**. **Focal vs nonfocal positive margin cut-off: 3 mm.**
- **Intraprostatic incision** (not "capsular incision" — there is no capsule): surgical transection of tumour; **progression risk equals focal EPE with a positive margin.**
- **Independent predictors of biochemical progression:** Gleason grade, EPE (focal vs nonfocal), margins (extent and grade), seminal vesicle invasion, nodal metastasis.

Tumour volume.

- **Not independently prognostic** after grade, stage and margins; reported only to match other organs → **rough estimate** (% of gland or maximum dimension) suffices.

Perineural and vascular invasion.

PERINEURAL INVASION — BIOPSY VERSUS RADICAL PROSTATECTOMY

Needle biopsy	Radical prostatectomy
Prognostic — report	Very common, not prognostic — omit

- **Vascular invasion** at prostatectomy is **uncommon** (7% of tumours <4 cm³; most modern tumours are <2 cm³); correlates with EPE, grade, margins and volume, yet **adds independent prognostic information**.

Concept — At prostatectomy the report is built from features that **independently predict progression** — grade, EPE extent, margin extent and grade, seminal vesicles and nodes; features that merely travel with these (volume, perineural invasion) are **demoted or dropped**.

Adenocarcinoma With Treatment Effect

GRADING AFTER TREATMENT

Treatment	Histological effect	Assign Gleason score?
Hormone therapy	Grade appears artefactually higher	No in areas with treatment effect; yes in areas without

Treatment	Histological effect	Assign Gleason score?
Finasteride	No consistent change in benign or malignant tissue	Yes
Radiotherapy	None, or radiation damage; benign glands mimic cancer (atypia up to 72 months)	Yes if no treatment effect; no if treatment effect
Cryotherapy / HIFU	Infarction of benign and neoplastic tissue; residual cancer shows no treatment effect	Yes

- **Post-radiotherapy biopsy — three diagnoses:** benign with radiation effect → cancer without treatment effect → cancer with treatment effect, the last carrying a **prognosis equivalent to no cancer**.
- **Tell the pathologist about prior radiotherapy. Trap:** ablated residual cancer **is** graded; hormone-treated cancer **is not**.

Concept — *Gleason score is only valid on untreated-looking architecture: where therapy has altered morphology the grade is not assigned; where therapy kills tissue outright (ablation), the survivors are unaltered and can be graded.*

SUBTYPES OF PROSTATE ADENOCARCINOMA

Mucinous adenocarcinoma.

- **Not more aggressive** than acinar cancer after radical prostatectomy.

Neuroendocrine differentiation and small cell carcinoma.

- **Immunohistochemical neuroendocrine differentiation** in acinar cancer risks **misdiagnosis as small cell carcinoma**.
- **Small cell carcinoma:** often **identical to lung small cell**; overlaps high-grade acinar carcinoma through **transdifferentiation**.

Ductal adenocarcinoma.

- **Large periurethral ducts** → **exophytic urethral lesion near the verumontanum** → obstruction or **haematuria**, mimicking **papillary urothelial carcinoma**. **Peripheral ducts** → presents like acinar cancer on needle biopsy.
- **Clinically underestimated:** DRE and PSA may be normal.
- **Pattern 4** because cribriform, with prognosis like acinar Gleason 8; whether graded ductal equals comparably graded acinar cancer is disputed.
- **After definitive therapy**, systemic agents effective in high-risk acinar disease have **limited use**.

Squamous and sarcomatoid carcinoma.

- **Squamous differentiation** is commoner in **hormone-treated adenocarcinoma** than as a pure primary. Sarcomatoid carcinoma = **carcinosarcoma**.

SUBTYPES COMPARED

Subtype	Frequency	Gleason grading	Behaviour / signature
Mucinous	—	Graded as usual	As acinar, incl. bone metastases
Neuroendocrine differentiation (immunohistochemical only)	Almost half of acinar	Graded as usual	No prognostic significance
Small cell	About 50% mixed	Not graded	Survival <1 year; pure = mixed
Ductal	Pure 0.4–0.8%; mixed about 5%	Pattern 4 (PIN-like 3; comedonecrosis 5)	Visceral recurrence at low PSA

Subtype	Frequency	Gleason grading	Behaviour / signature
Squamous (pure)	Rare	—	Osteolytic metastases; hormone-unresponsive; poor survival
Sarcomatoid	Rare	—	Dismal

- **Pattern:** morphology that resembles acinar cancer (mucinous, immunohistochemical neuroendocrine) behaves like it; departures (small cell, squamous, sarcomatoid) escape androgen control and do badly.
- Trap:** ductal is pattern 4 by default, not 3.

Urothelial Carcinoma

- **Prostatic duct and acinar involvement** in patients with **bladder carcinoma in situ (CIS)** treated for months to years with intravesical therapy; **35–45% of cystoprostatectomies** for urothelial carcinoma show prostatic involvement.
- **Intraductal urothelial carcinoma only** → **does not worsen prognosis**; bladder stage decides.

STAGING OF UROTHELIAL CARCINOMA IN THE PROSTATE

Route	Finding	Stage
From CIS of the prostatic urethra (bladder staged separately)	Invasion of suburethral tissue	pT1
	CIS in prostatic acini with prostatic stromal invasion	pT2
Direct from a bladder tumour	Transmural invasion into prostatic stroma	pT4

POORLY DIFFERENTIATED PROSTATIC VERSUS UROTHELIAL CARCINOMA — IMMUNOHISTOCHEMISTRY

Prostatic adenocarcinoma	Urothelial carcinoma
PSA — most positive, may be focal	GATA3 — most sensitive and specific (some prostate cancers aberrantly positive)
If PSA weak or negative: P501S (prostein) , NKX3.1	Less sensitive: uroplakin II, thrombomodulin, p63, HMWCK

- **Trap:** prostatic stromal invasion is **pT2 from urethral CIS but pT4 from the bladder**.

Mesenchymal Tumours

- **Sarcomas = 0.1–0.2%** of prostatic malignancy.
- **Rhabdomyosarcoma: commonest in childhood** (rare in adults). **Leiomyosarcoma: commonest adult sarcoma**.
- **Inflammatory myofibroblastic tumour:** any age, may follow TUR (or not); **mimics leiomyosarcoma**.
- **Specialised prostatic stromal tumours:** range from **stromal tumour of uncertain malignant potential (STUMP)** to **stromal sarcoma**; one variant resembles breast **phyllodes tumour**.
 - **STUMP:** usually indolent, but may recur rapidly, coexist with or **progress to stromal sarcoma** → **close follow-up; consider definitive resection in younger patients**.
 - **Stromal sarcoma:** can metastasise.

Miscellaneous Malignant Tumours

- **Secondary lymphomatous infiltration is far commoner** than primary prostatic lymphoma.
- **Commonest leukaemic involvement: chronic lymphocytic leukaemia/small lymphocytic lymphoma;** monocytic, granulocytic and lymphoblastic forms also occur.

KEY POINTS

- **LGPIN not reported** (irreproducible, clinically insignificant); **isolated HGPIN, even multifocal, does not justify repeat biopsy** once DRE and PSA are considered.
- **IDC-P: usually spread of high-grade invasive carcinoma**; only a minority are high-grade precursors.
- **Atypical biopsy** → about **40% cancer** regardless of PSA; 80% grade group 1 favours markers, imaging and selective targeted re-biopsy, while the 20% of higher grade lead some to re-biopsy routinely.
- **Atypical cases change most often on expert review** → consult before re-biopsy.
- **Prostatectomy tumour volume** is not independently prognostic after grade, stage and margins.
- **Gleason grade**, on any specimen, remains among the most influential prognostic factors; **grade groups** are simpler, more intuitive and more prognostic.